# PRC-DB-007 - orders-db failover

PostgreSQL primary to standby failover for the orders-db cluster

| Field | Value |
|---|---|
| Owner | Amara Teodorescu, Database Administrator |
| Approved by | Jun Hallgarth, Engineering Manager |
| Effective | 2026-02-23 |
| Scope | orders-db: primary in zone 1, synchronous standby in zone 2, asynchronous read replica for reporting-batch |
| Targets | RPO 0 with the synchronous standby; RTO 15 minutes |

## 1. When to use

Use this procedure when the orders-db primary is unavailable or degraded and the incident commander of a SEV1 or SEV2 incident has decided to fail over. Do not fail over for a single slow query.

## 2. Preconditions

- An incident commander is assigned and the incident channel is open.
- Replication lag of the standby is below 30 seconds (unless the primary is lost).
- Deployments to checkout-web and ticketing-api are frozen.

## 3. Steps

1. Announce the failover in the incident channel with the expected impact.
2. Check the replication lag on the standby and record it.
3. Stop writes: scale the checkout-web order workers to 0 replicas.
4. Promote the standby to primary.
5. Point the DNS alias orders-db-rw to the new primary (TTL 30 s).
6. Restart the ticketing-api pods to drop pooled connections.
7. Verify writes with a synthetic test order.
8. Scale the checkout-web order workers back to their normal count.
9. Rebuild the old primary as the new standby.
10. Confirm that monitoring shows replication healthy again.
11. Hand over to the on-call engineer and update the incident timeline.
12. Hold the postmortem within 5 working days.

## 4. Rollback

If the promoted standby fails verification, stop and escalate to the database administrator on call. Do not promote the asynchronous read replica: it may miss committed orders.

## 5. Verification checklist

| Check | Expected |
|---|---|
| orders-db-rw resolves to the new primary | yes |
| Synthetic order written and read back | yes |
| checkout-web error rate | below 1 % |
| Replication to the new standby | lag below 30 s |